

Tool Documentation – Windows11LiveDataAcquisitionToolkit

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Document Revision History

Name	Revision Date	Version	Description
Tytus Gonzalez	2026-03-25	0.1	Draft
Tytus Gonzalez	2026-03-26	1.0	Final

Purpose

The purpose of the Windows Live Data Acquisition Toolkit is to collect volatile and system-related information from a live Windows 10/11 machine for forensics analysis. This tool helps investigators quickly gather critical data during an active investigation without running multiple commands. This automates the process into a single script to run and ensures consistency, efficiency, and performance.

The toolkit captures a lot of system and network artifacts such as system information, running processes/services, network traffic, active connections, and routing tables. These artifacts are very important for identifying suspicious activity, malware, or unauthorized access.

The tool will execute several commands and afterward, it generates multiple output files in different formats such as txt, CSV, and JSON. The tool will also create a SHA256 hash file that contains hash values of the outputs to ensure integrity of the data collected.

Legal Notice

This tool is intended for educational and authorized forensics use only. It should only be executed on systems that are explicitly listed for a forensics case. Unauthorized use of this tool on systems without proper consent may violate local, state, or federal laws.

The author is not responsible for any misuse, damage, or legal consequences resulting from the use of this tool. Users are responsible for ensuring they comply with all applicable laws, organizational policies, and ethical guidelines when conducting forensic investigations.

By using this tool, the user acknowledges that they have proper authorization and accept full responsibility for its use.

Pre-requisites

Before running this tool, the following requirements must be met:

- Windows 10 or Windows 11 Operating System
- Windows PowerShell 5.1 or later version
- The user must have authorized access to the system
- The script must be executed with Administrator privileges to ensure access to all system and network information is available
- PowerShell script execution must be enabled
- A script editor is recommended for editing and running the script (e.g., VSCode)

Tool Usage

The Windows Toolkit is executed using Windows PowerShell and requires administrative privileges to function properly.

Step 1: Open PowerShell as Administrator

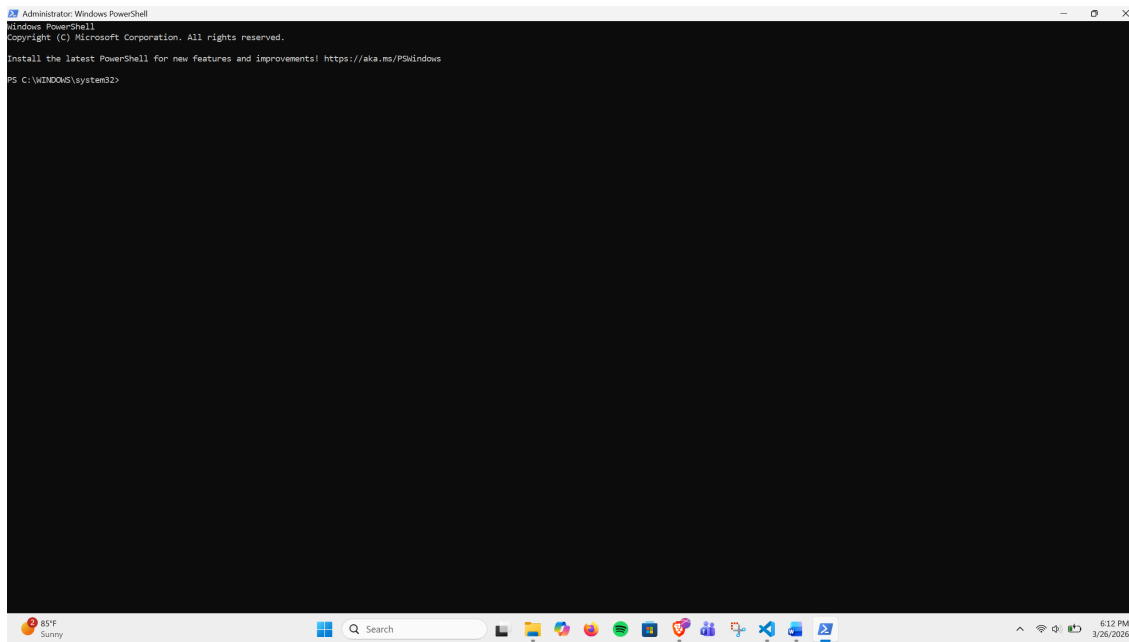


Figure 1: Open PowerShell as Administrator

Step 2: Navigate to the script's location

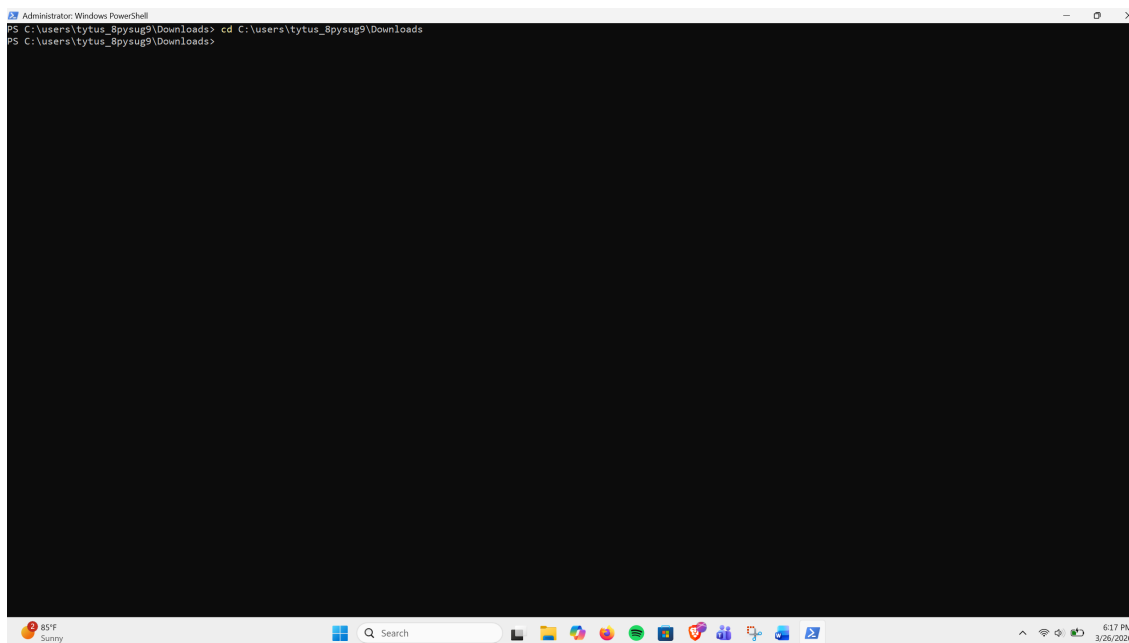


Figure 2: Navigating to my scripts folder in PowerShell

Step 3: Enable script execution

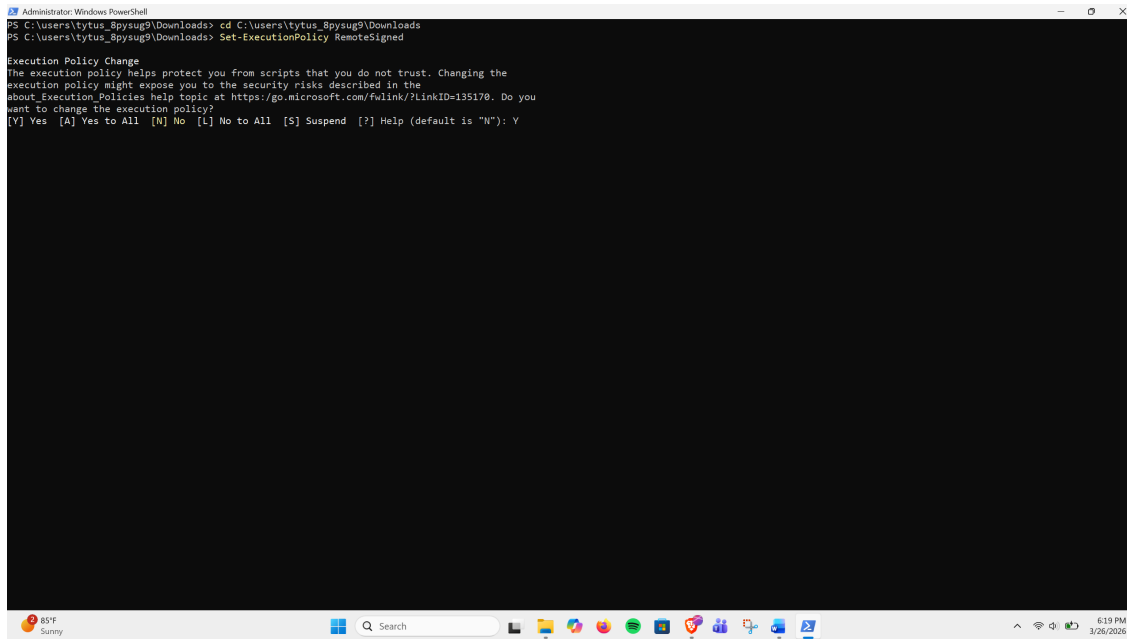


Figure 3: Enable script execution

Step 4: Run the script

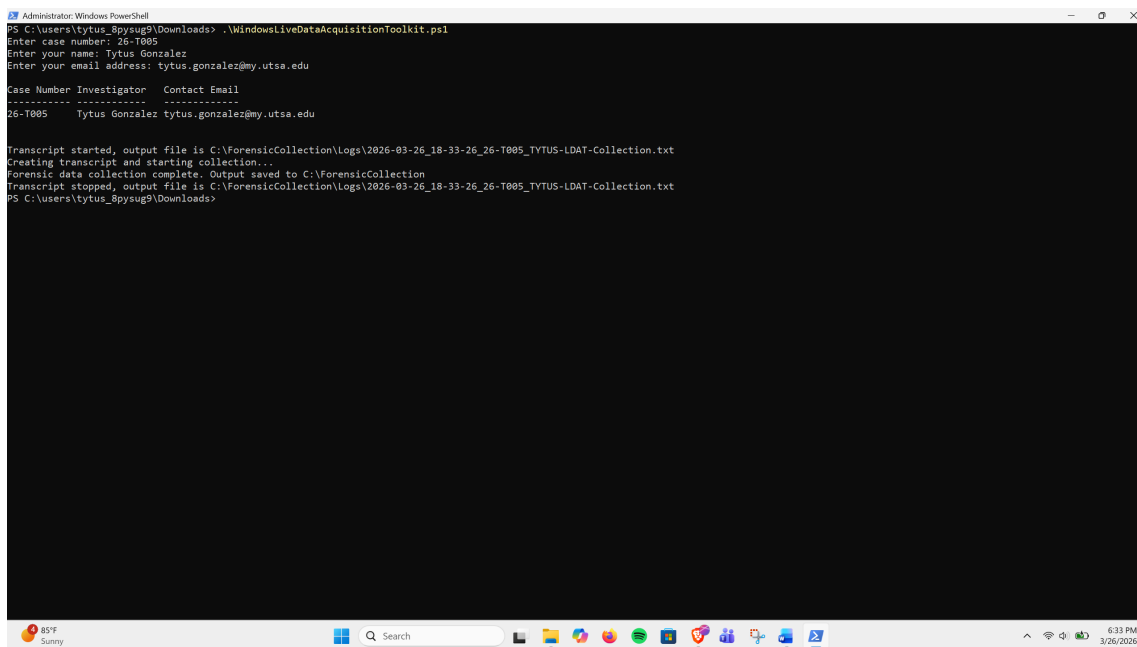
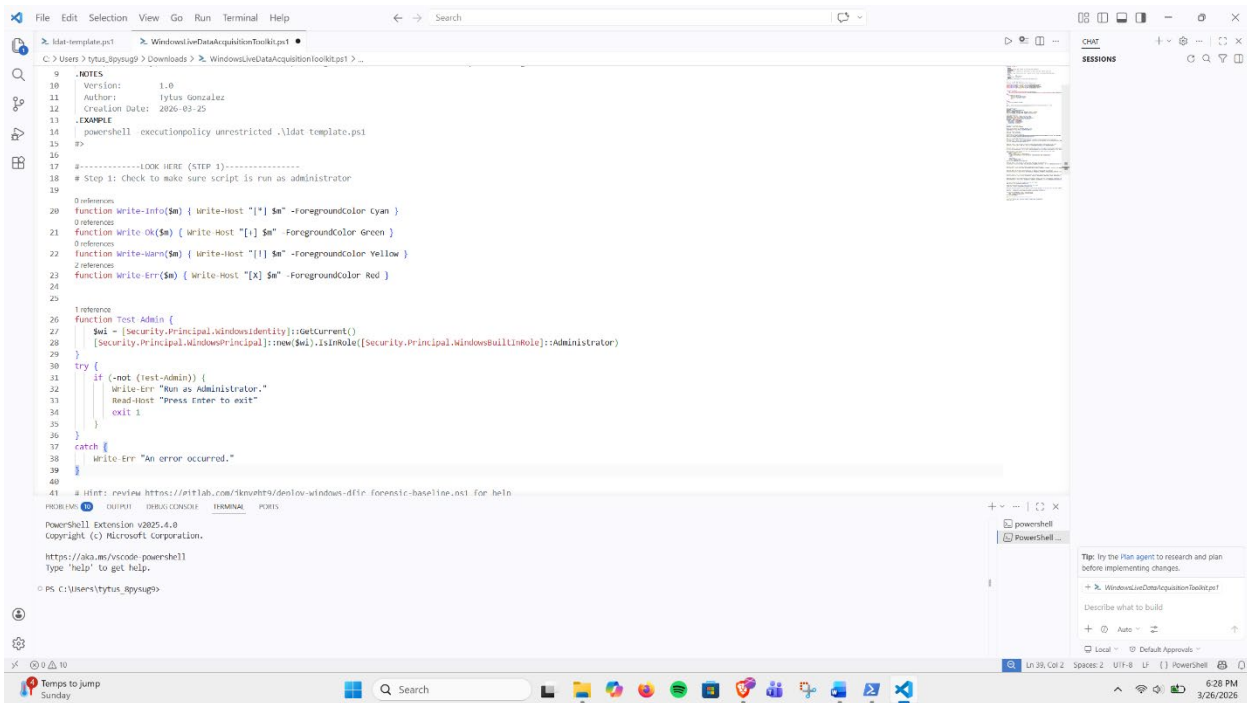
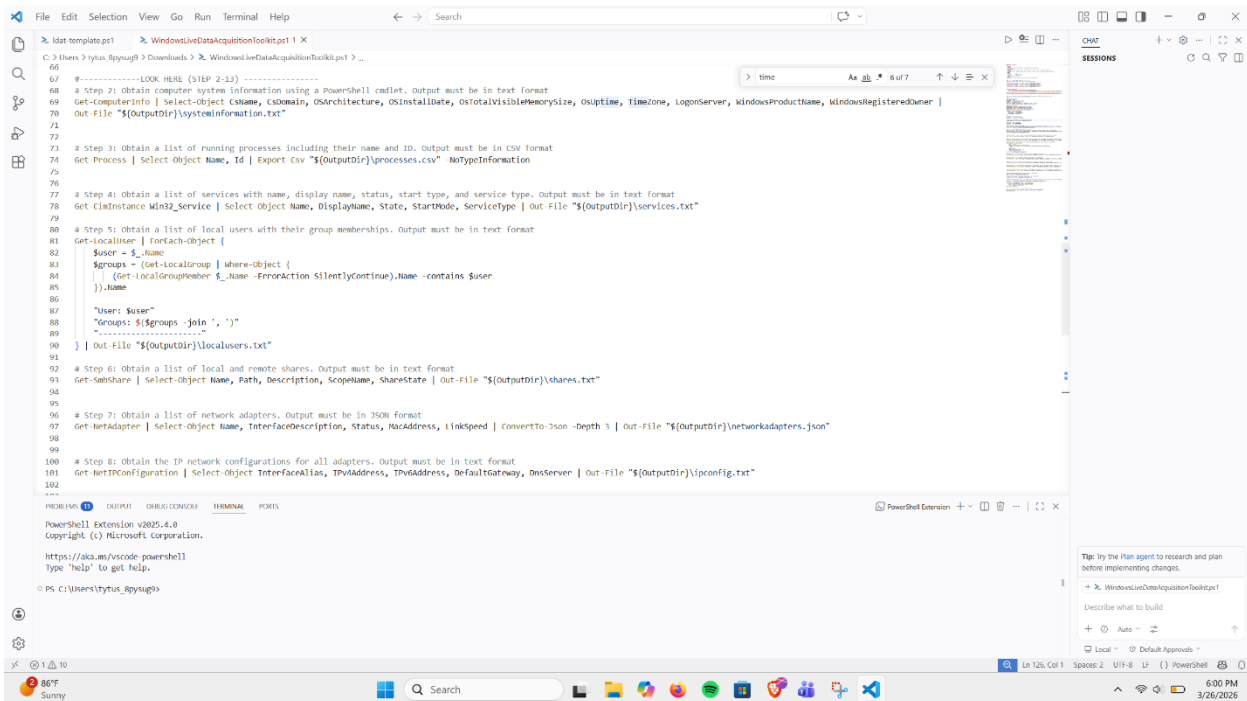


Figure 4: Script execution and user input prompts

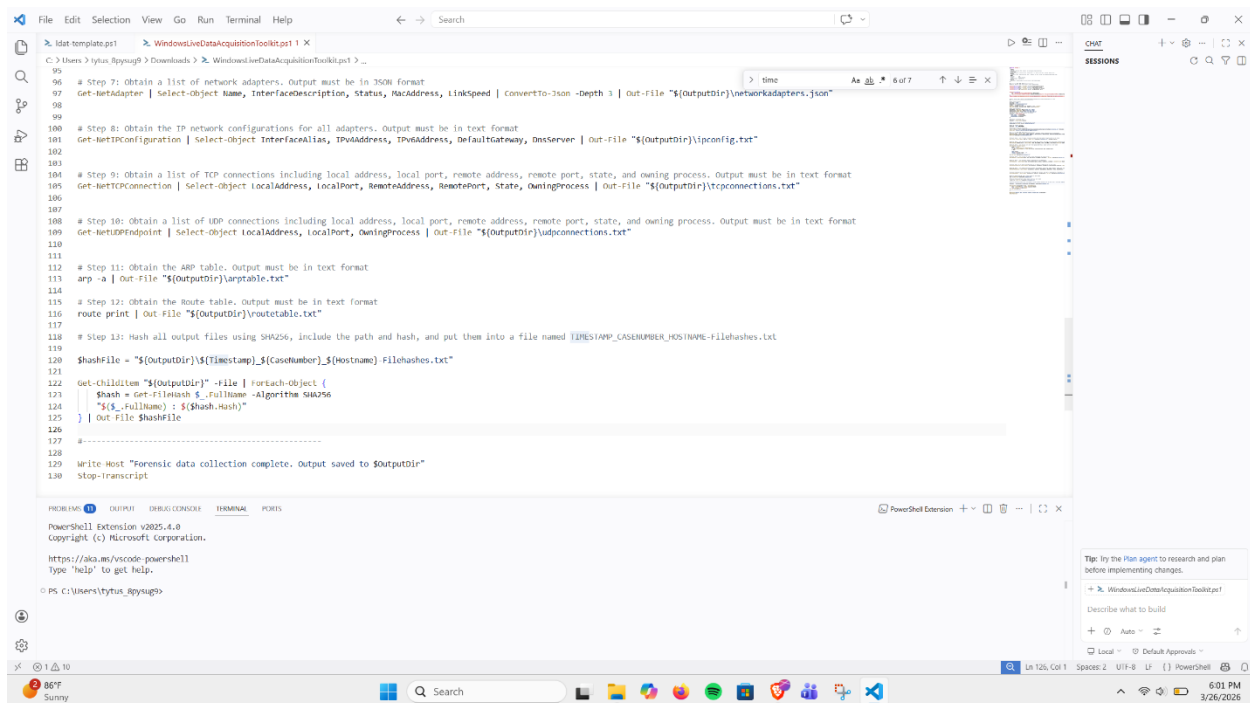
1. Administrator Privileges Check- This step checks if the script is running with administrator privileges. This ensures all system data can be collected properly.
2. System Information- This step collects basic system information such as OS details, hostname, and memory to establish a system baseline.
3. Running Processes- This step gathers all running processes with their names and IDs to help identify suspicious programs.
4. Services- This step collects information about system services, including their status and startup type, to detect abnormal services.
5. Local Users and Groups- This step lists local users and their group memberships to identify unauthorized or privileged accounts.
6. Shared Resources- This step lists shared folders and resources that may allow access to files on the system.
7. Network Adapters- This step collects information about network adapters, including status and MAC address, to understand connectivity.
8. IP Configuration- This step gathers IP addresses, gateways, and DNS information to show how the system connects to the network.
9. TCP Connections- This step lists active TCP connections to identify ongoing network communication.
10. UDP Connections- This step collects UDP endpoints to help detect unusual network activity.
11. ARP Table- This step displays IP-to-MAC address mappings to identify devices on the local network.
12. Route Table- This step shows the system's routing table to understand how network traffic is directed.
13. File Hashing- This step generates SHA256 hashes of all output files to verify data integrity.



Step 1 Script in VSCode editor



Step 2-8 Script in VSCode editor



Step 9-13 Script in VSCode editor

Output Details

This tool generates multiple output files containing collected forensic data from the system and the network. All output files are stored in the directory: C:\ForensicCollection

Each file belongs to a specific category of system information:

- systeminformation.txt- contains general system details such as OS, version, hostname, and memory
- processes.csv- lists all running processes with their names and process IDs
- services.txt- displays system services, including their status and startup type
- localusers.txt- shows local user accounts and their group memberships
- shares.txt- lists shared resources on the system
- networkadapters.json- contain details about network interfaces in JSON format
- ipconfig.txt- displays IP configuration for all network adapters
- tcpconnections.txt- lists active TCP network connections
- udpconnections.txt- lists active UDP endpoints
- arptable.txt- shows the system's routing table
- TIMESTAMP_CASENUMBER_HOSTNAME-Filehashes.txt- contains SHA256 hashes of all output files for integrity verification

To read the output files, the user must:

- .txt files can be opened with Notepad or a text editor
- .csv files can be opened with Excel
- .json files can be opened with VSCode

Side Effects

What are the side effects to your tool. How does it affect the target system and possibly damage, destroy, overwrite, or add evidence.

This tool has little effect on the target system because it mostly performs read-only functions to collect information. However, there are a few side effects such as:

- The tool creates new files and stores all the files in a new directory
C:\ForensicsCollection
- Running the tool may cause slight system impact due to the execution of multiple commands
- The script accesses sensitive system information which could be logged or cached by the operating system

The tool does not modify, delete, or overwrite existing system data. Investigators should be aware that any live data acquisition tool can slightly alter the system state and they should document its use.

Contact Information

In the case of questions or issues to report, please contact the following individual or organization.

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